2024

# SHARED VISION & UNIFIED STRATEGY:

## ENHANCING PUBLIC HEALTH INFRASTRUCTURE THROUGH BI-DIRECTIONAL RELATIONSHIPS IN PRACTICE AND ACADEMIA

ASPPH ASSOCIATION OF SCHOOLS & PROGRAMS OF PUBLIC HEALTH
THE VOICE OF ACADEMIC PUBLIC HEALTH

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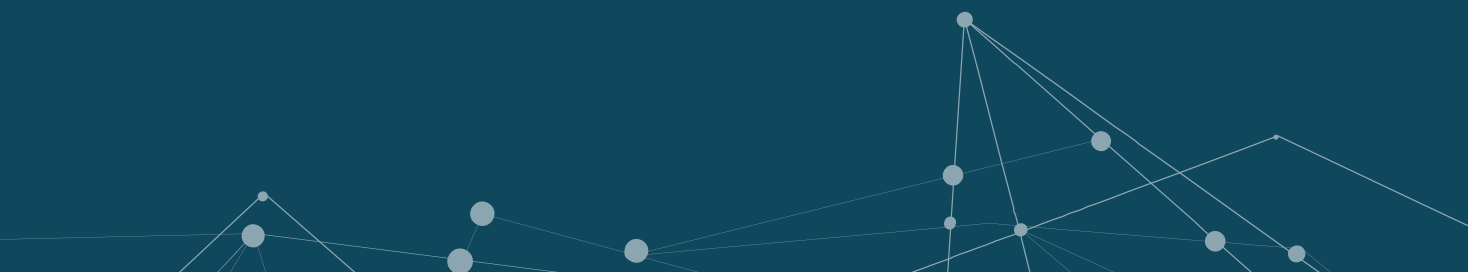
# EXECUTIVE SUMMARY

The COVID-19 pandemic revealed significant challenges within the United States public health system, highlighting deficiencies in funding, infrastructure, and data systems. The challenges emphasized the necessity for collaboration and bi-directional relationships between governmental public health practice and academia to address public health emergencies effectively. The Association of Schools and Programs of Public Health (ASPPH) and its members recognize this need and are committed to fostering collaborative, bi-directional relationships to enhance public health response and infrastructure. As such, ASPPH has partnered with the National Network of Public Health Institutes (NNPHI) to explore successes, challenges, and opportunities to leverage and enhance bi-directional relationships between governmental public health practice and academia.

To supplement a review of the literature, focus groups were conducted with health departments in the U.S. Northeast, South, and Midwest. These focus groups revealed variations in the depth and impact of bi-directional relationships. They identified mutual benefits, such as increased research capacity and credibility, but also highlighted challenges like resource limitations and sustained, long-term engagement. This report highlights the findings of both the literature review and focus groups, and proposes a framework for successful bi-directional relationships along with actionable recommendations for the various public health organizations, agencies, and institutions with which success lies.

This report is intended to spark discussion. Your feedback is welcome and wanted as this serves as just one step toward enhancing public health infrastructure through bi-directional relationships in practice and academia.



CENTER FOR PUBLIC HEALTH WORKFORCE DEVELOPMENT ASPPH

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# EXECUTIVE SUMMARY
## *(CONTINUED)*

**Key Takeaways:**

"A bi-directional relationship is defined as the movement or sharing of information between the community and the academic institution in a manner that is mutually beneficial (Haynes, et al. 2011, p. 1365)". Bi-directional relationships of trust and collaboration enable partners to identify and address health disparities, building the capacity to assess and address community needs and develop effective, sustainable interventions and programming (Zimmerman et al., 2018).

Governmental public health practice and academia are strong partners with significant opportunity to expand partnership for technical assistance, workforce development, workforce recruitment and retention, and student training.

To achieve successful and sustained bi-directional relationships, governmental public health agencies and schools and programs of public health must develop relational foundations based in a shared vision, collaborative agenda, and integrated resources. This includes defining a formal or informal structure for partnership, including goals and objectives, roles and responsibilities, integration of resources, organizational capacity, and consistent communication.

Bi-directional relationships between governmental public health agencies and academic institutions are essential for a resilient public health system. These partnerships not only address immediate public health needs but also build a foundation for future challenges.

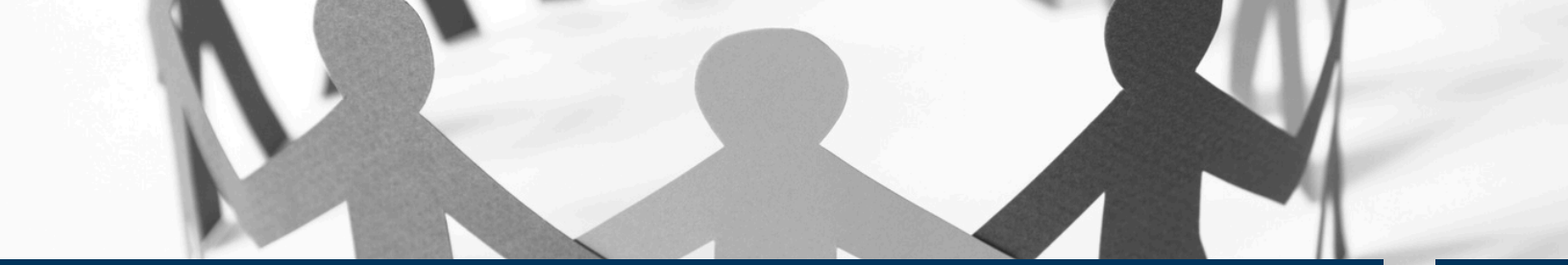
# INTRODUCTION

The COVID-19 pandemic revealed significant obstacles impacting the ability of the nation's public health system to respond to emergencies. In addition to a lack of sustained and adequate funding for governmental public health agencies, there is a clear need to strengthen public health infrastructure, modernize public health data systems, and eliminate gaps resulting in health disparities. The public health system stands in a pivotal place where partnerships and bi-directional relationships between governmental public health agencies and academic institutions are critical to deliver essential services and enhance response to emergencies.

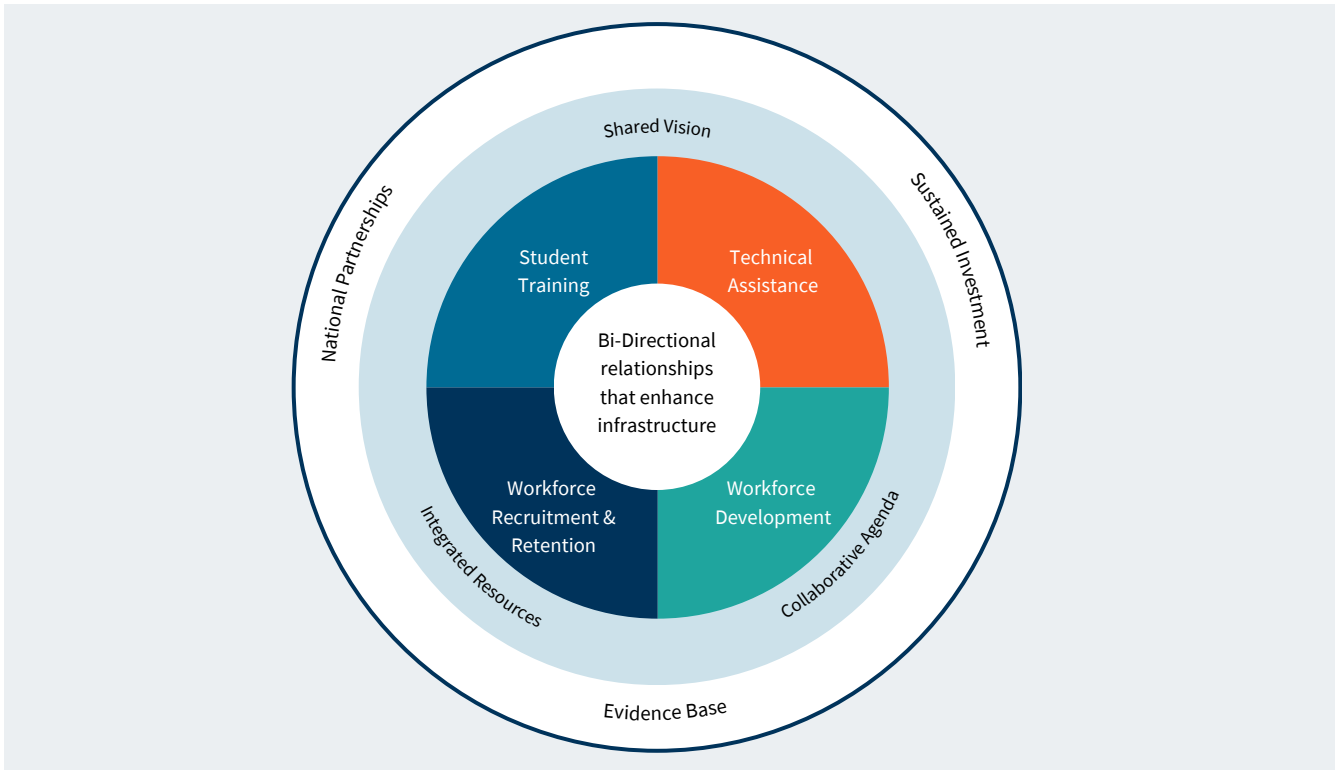
The pandemic also revealed the critical role of the governmental public health workforce. In support of this workforce, the Association of Schools and Programs of Public Health (ASPPH) and our member schools and programs of public health have a renewed interest and call to action to join our practice partners and colleagues at the federal, state, Tribal, local, and territorial levels in collaborative partnerships and bi-directional relationships.

ASPPH represents 150 graduate schools and programs of public health accredited by the Council on Education for Public Health. In 2022, these schools and programs of public health enrolled over 80,000 students and employed over 10,000 faculty and administrators, many of whom engage governmental public health agencies and their constituents. In collaboration with the National Network of Public Health Institutes (NNPHI), ASPPH set out to characterize bi-directional relationships between governmental public health practice and academia, thereby exploring benefits, challenges, and opportunities of bi-directional relationships.

***With this exploration, ASPPH proposes recommendations for developing and/or sustaining bi-directional relationships between governmental public health practice and academia to improve public health infrastructure.***



# KEY RECOMMENDATIONS




National public health organizations and agencies must partner to champion sustained investment in public health infrastructure *and* the ongoing development of the evidence base around and facilitation of bi-directional relationships between governmental public health practice and academia.

In collaboration with governmental public health agencies, including state, Tribal, local and territorial health departments, schools and programs of public health must initiate and foster the development and implementation of a shared vision and collaborative agenda that is supported by integrated resources.

In direct consultation with governmental public health agencies, including state, Tribal, local and territorial health departments, schools and programs of public health must prioritize and advance efforts around technical assistance, workforce development, workforce recruitment and retention, student training, and other to-be-identified needs.

# DEFINING BI-DIRECTIONAL RELATIONSHIPS

## DEFINITION

"A bi-directional relationship is defined as the movement or sharing of information between the community and the academic institution in a manner that is mutually beneficial (Haynes, et al. 2011, p. 1365)". Bi-directional relationships of trust and collaboration enable partners to identify and address health disparities, building the capacity to assess and address community needs and develop effective, sustainable interventions and programming (Zimmerman et al., 2018). They ensure that partners define responsibilities and establish shared goals (Mabachi & Kimminau, 2012). "Bi-directional" means that communication flows in both directions and all partners have some share of ownership, decision-making, monitoring and revision, and promotion of the partnership model (Baquet, 2012, p. 1807).

Bi-directional relationships are fundamental to successful public health practice and ensure a competent and skilled public health workforce (Petroro et al., 2011). Further, as evidenced during the COVID-19 early response, bi-directional relationships between governmental public health agencies and schools and programs of public health are vital in effectively addressing public health emergencies (Lederer et al., 2022; Hamilton et al., 2014). While governmental public health agencies and academia sometimes struggle to develop and maintain bi-directional relationships, each has much to offer the other, particularly with respect to teaching, research, service, and governance (Akintobi et al., 2014). Bi-directional relationships are an important means for advancing evidence-based decision-making and for implementing evidence-based programs and policies. We must consider how bi-directional relationships are leveraged to address current and future public health issues and how such partnerships will influence the forces of change likely to be experienced in public health practice (Erwin et al., 209 and Erwin, Harris, et al., 2016).

## METHODOLOGY

This report highlights the findings from both a review of the literature and focus groups with governmental public health agencies to characterize bi-directional relationships between governmental public health practice and academia and to identify opportunities to establish, expand, and foster these relationships and strengthen public health infrastructure.

Using search engines and databases, including PubMed, Google Scholar, and Google, researchers conducted a review of the literature. Articles were identified from peer-reviewed journals and from gray literature, published in English, within the lens of academic and practice-based public health and health collaboration in the United States between the years of 2011 and 2024.

Search terms included: *public health capacity building, public health program facilitation, public health workforce recruitment, public health workforce development, mentorship in public health, academic health department, public health practice-based education, public health practice-based training, public health competency development, dean's advisory council, public health journal club, public health grand rounds, bi-directional partnerships, bi-directional relationships, health department academic partnerships, ASTHO practice and academic partnerships, public health department academic partnerships, local health department academic partnerships, public health education capacity building, and public health training.*

Researchers reviewed website program description sections, article titles, and abstracts to identify sources that describe relationships in governmental public health practice and academia. Researchers completed a comprehensive review of each source, identifying descriptions and examples of bi-directional relationships between governmental public health agencies and schools and programs of public health. In addition, researchers reviewed previously collected ASPPH survey data relevant to the topic of bi-directional relationships

Literature review findings informed the execution of a series of focus groups with governmental public health agencies. Researchers conducted focus groups with three health departments in the United States, including:

| | |
|---|---|
| **State Health Department** | A state health department in the Northeast, serving a population of approximately 3.6 million. |
| **Large Local Health Department** | A county health department in the South, serving a population of approximately 2.6 million. |
| **Mid-Sized Local Health Department** | A county health department in the Midwest, serving a population of approximately 600,000 people. |

Researchers invited health department leaders and staff, including health department directors, workforce development directors, staff involved in internship and practicum placements, and more. Focus groups followed a semi-structured format, using the following research questions:

1. What is an academic/practice relationship?
2. What are examples of these relationships at each health department?
3. How do health departments establish these relationships?
4. What is the structure of these relationships?
5. What are the benefits of these relationships? What are some successes?
6. What are the challenges of these relationships?
7. How do health departments maintain these relationships?

Focus groups were conducted on-site and in-person at all three governmental public health agencies. Researchers recorded the 90-minute focus groups, and recordings were transcribed by an external transcription service. Researchers analyzed the focus group transcriptions using Dedoose qualitative software and a code book framed around the research questions (Dedoose, 2021).

# CHARACTERIZING BI-DIRECTIONAL RELATIONSHIPS

## *ACADEMIC HEALTH DEPARTMENTS*

An academic public health department, or academic health department (AHD), is defined as "an arrangement between an academic institution and a governmental public health agency that provides mutual benefits in teaching, research, and service, with academia informing the practice of public health, and the governmental public health agency informing the academic program" (Erwin & Keck, 2014, p.270). The academic institution may be a school or program focused on a health profession (e.g., a school of medicine, nursing, or public health), a department within a school, or a community college that is building practice partnerships for future employment opportunities. The public health agency is usually a state or local health department, but may also be a Tribal, territorial, or federal public health agency. The relationship may range from informal and passive to formal, active partnerships with a signed memorandum of understanding (MOU) that outlines the sharing of financial, human, and physical resources (Erwin et al., 2022).

AHDs are designed to provide collaborative opportunities across governmental public health practice and academia, involving field practice for students, practice-based research, and public health practice workforce development, resulting in practice-informed teaching and academic-informed practice (Erwin et al., 2022). The AHD structure provides opportunities for schools and programs of public health faculty and governmental public health agency staff to share in leadership on grant-supported projects or research, to develop adjunct faculty appointments, to lead intentional efforts to build a diverse and inclusive workforce, and other mentorship/leadership roles (Ramos et al., 2022).

The term "academic health department" dates back to the year 2000, although there are some examples of governmental public health practice and academic partnerships before that date (Erwin et al., 2022). Since 2000, there has been growing consensus around the mutual value of AHDs. The Council on Education for Public Health (CEPH) now requires that institutions provide evidence of practice engagement as part of their academic accreditation process.

Further, the Public Health Accreditation Board's (PHAB) program for national accreditation includes metrics for examining the extent to which health departments are working in collaboration with academic programs to strengthen the workforce and identify the best evidence for making informed public health practice decisions (Erwin, Harris, et al., 2016).

In 2015, to continue developing the evidence base for AHDs and to provide a better estimate of the prevalence of AHDs, researchers surveyed CEPH-accredited schools and programs of public health. The survey asked questions about partnership agreements, functions of the AHD, and experienced and potential benefits of the AHD. It also included questions regarding public health practice partners and the various types of engagement activities, as well as questions on PHAB accreditation-related activities (Erwin, Harris, et al., 2016).

Of the 117 institutions who responded to the survey, 55% reported having an AHD with the majority of these having a formal written agreement. These institutions reported various forms of engagement with health departments. For example, more than 70% of the AHD respondents indicated that their governmental public health practice partner actively recruited their graduates for employment and that academicians reported substantial involvement in the prerequisites for accreditation through PHAB (Erwin, Harris, et al., 2016).

According to the Public Health Foundation, as of June 2024, there are AHDs in 32 states. However, since there is no formal designation of AHDs, they are likely undercounted. Though AHDs themselves may be undercounted, they are recognized as the most robust and established form of partnership between governmental public health practice and academia. As such, perspectives from governmental public health agencies with lower levels of collaboration have largely been left undefined and under-examined (Kovach et al., 2019b).

Some governmental public health practice and academic partnerships, while not labeled as AHDs, have functioned in a similar capacity (Erwin et al., 2022):

**Prevention Research Centers (PRCs):** Authorized by Congress in 1984 and funded and managed through the CDC, PRCs are academic research centers established in either accredited schools of public health or schools of medicine with a preventive medicine residency program. Their function is to study how people and their communities can reduce the risks for chronic illnesses.

**Public Health Training Centers (PHTCs):** Established by the Health Resources and Service Administration (HRSA) in 1999 to provide training and expand competencies of the public health workforce at the state, local, and Tribal levels.

**Preparedness and Emergency Response Learning Centers (PERLCs)** and **Preparedness and Emergency Response Research Centers (PERRCs):** Academic Centers for Public Health Preparedness were established by the CDC in 2000 and subsequently replaced by PERLCs and PERRCs. PERRCs were established to support applied public health systems research to strengthen and improve national public health preparedness and emergency response capabilities, while PERLCs aim to enhance the public health workforce competencies in emergency preparedness and response.

**Regional Medicine-Public Health Education Centers (RM-PHECs):** Funded by the CDC and managed through the Association of American Medical Colleges to improve integration of public health, population health, and prevention education into medical school curricula and to develop new field opportunities for medical students at public health agencies.

**Public Health Practice-Based Research Network (PBRN):** Funded by the Robert Wood Johnson Foundation from 2007-2015, this initiative supported teams of academicians and practitioners to conduct research on the organizations, financing, and delivery of public health services and to apply research findings to improve public health practice and policy. As of 2013, 926 local health departments, 20 state agencies, and 35 academic units had been involved in PBRN-funded studies.

All three health departments that participated in ASPPH focus groups have at least one AHD relationship.

**FOCUS GROUP FINDINGS**

**State Health Department** is currently developing an AHD with the state community college system and is already working with this academic institution to develop an associate's degree in public health.

**Large Local Health Department** has one active, formal AHD and is currently developing a MOU with another academic institution to form its second AHD partnership.

**Mid-Sized Local Health Department** has at least one active AHD partnership in which academic institution faculty support the health department with strategic planning and health department staff serve as adjunct faculty.

# BI-DIRECTIONAL RELATIONSHIPS

## ACTIVITIES

Relationships, both formal and informal, often exist between governmental public health agencies and academic institutions outside of an AHD framework. These relationships are often formed around shared interests and priorities. The 2022 National Profile of Local Health Departments from the National Association of County and City Health Officials (NACCHO) found that 89% of local health departments work with colleges or universities “in any way”, and 64% regularly schedule meetings, have written agreements, or share personnel/resources with colleges or universities.

Further exploring these relationships, ASPPH conducted a survey of its member schools and programs of public health in 2023 to learn more about formal and informal relationships with state, Tribal, local, and territorial health departments. The survey was conducted to inform the work of the ASPPH Public Health Workforce Development Taskforce and CDC Public Health Leadership and Education, Advancing Health Equity and Data Science (PH LEADS) as it develops frameworks, recommendations, and resources to establish and/or grow more fruitful relationships with colleagues in governmental public health departments. The survey was designed with input from practice faculty at schools and programs of public health and identified several collaborative activities conducted through bi-directional relationships:

## Bi-Directional Relationship Activities

### TECHNICAL ASSISTANCE

- Facilitation (Strategic Planning, CHA/CHIP)
- Workforce evaluation and assessment
- Health department accreditation
- Research and grants collaborations

### WORKFORCE DEVELOPMENT

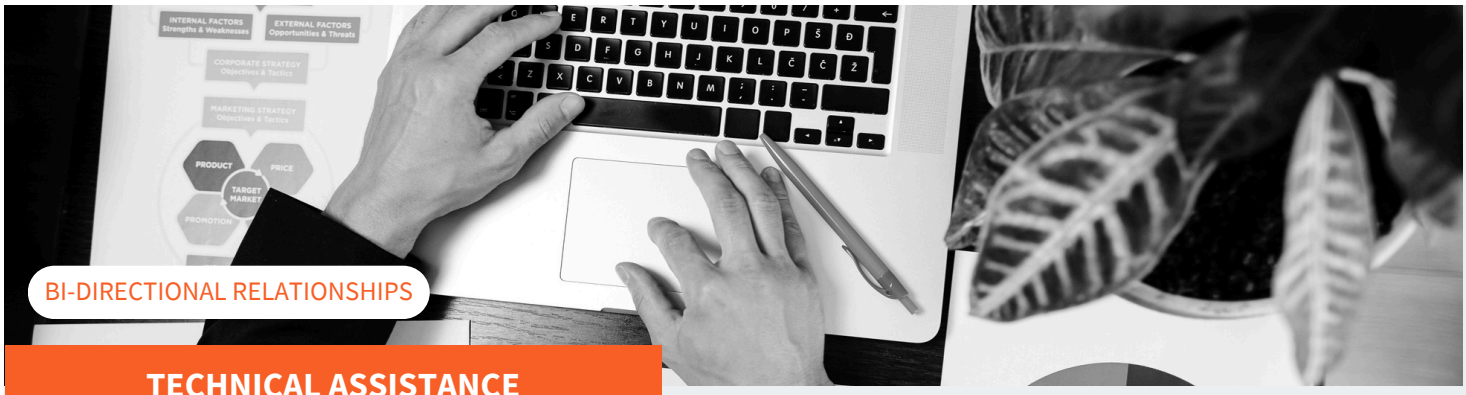
- Continuing education
- Knowledge/skills training
- Leadership development
- Executive coaching

### WORKFORCE RECRUITMENT AND RETENTION

- Shared staffing
- Adjunct faculty
- Surge capacity staffing
- Joint search committees
- Outsource hiring

### STUDENT TRAINING

- Practicum/internship
- Guest lectures
- Curriculum/advising
- Apprenticeship programs
- Field trips
- Journal clubs
- Grand rounds
- Shadowing/rotating through agency




BI-DIRECTIONAL RELATIONSHIPS

## TECHNICAL ASSISTANCE

ASPPH-member schools and programs of public health are collaborating on technical assistance priorities with state and local health departments, with research and grant collaborations being the most common type of relationship. Of the 35 ASPPH-member schools and programs of public health that participated in the 2023 survey, 94% are partnering on technical assistance activities. Specifically, 83% collaborated with a state and/or local health department on research and grants, 69% on facilitation of strategic planning, community health assessments (CHA) or community health improvement plans (CHIP), 46% on workforce evaluation and assessment, and only 14% collaborated on health department accreditation.

## TECHNICAL ASSISTANCE (CONTINED)

Survey results were reflected in the focus group findings.

### FOCUS GROUP FINDINGS

**State Health Department** partners with an academic institution and the CDC on a state-wide epidemiology program to assess the public health impact of emerging and re-emerging infections and to evaluate methods for their prevention and control.

**Large Local Health Department** works closely with an academic institution on its Community Health Improvement Plan and stated that academic institutions serve as the lead evaluators for many of the health department's grant-funded programs.

**Mid-Sized Local Health Department** works closely with its AHD-partner academic institution on strategic planning for the health department as a whole, including developing logic models for each department division and its respective programs. These partner institutions are currently planning a retreat in which they will select Key Performance Indicators for the health department. The Mid-Sized Local Health Department also has a Health Department Leadership Council on which one of its closest academic partners has a representative.

BI-DIRECTIONAL RELATIONSHIPS

## WORKFORCE DEVELOPMENT

ASPPH-member schools and programs of public health are collaborating on workforce development priorities with state and local health departments, with continuing education collaborations being the most common. Of the 35 ASPPH member schools and programs of public health that participated in the 2023 survey, 89% noted at least one collaborative relationship with a state and/or local health department on workforce development efforts. Specifically, 77% partnered with a state and/or local health department(s) on continuing education, 71% on knowledge/skills training, 34% on leadership development, and only 3% (one ASPPH member) on executive coaching.

### FOCUS GROUP FINDINGS

Despite survey results indicating high levels of workforce development collaboration with state and local health departments, the state health department was the only focus group that noted a workforce development relationship with a school or program of public health.

**State Health Department** is currently developing contracts with schools of public health for "workforce development" yet provided no additional information. In the past, however, an academic institution had curated courses for its health department workforce using CDC's TRAIN.

BI-DIRECTIONAL RELATIONSHIPS

## WORKFORCE RECRUITMENT AND RETENTION

ASPPH-member schools and programs of public health are collaborating on workforce recruitment and retention priorities with state and local health departments, though these types of partnerships were noted less frequently than technical assistance, workforce development, and student training priorities. Of the 35 ASPPH member schools and programs of public health that participated in the PH LEADS survey, 63% noted at least one collaborative relationship with a state and/or local health department on workforce recruitment and retention efforts. Regarding specific collaboration types, 51% had adjunct faculty from a state and/or local health department serving at their institution, 26% reported surge capacity staffing partnership with a health department(s), 20% shared staff with a health department(s), 14% had a joint search committee with a health department(s), and only 3% (one ASPPH member) noted working with a health department(s) on outsource hiring.

Despite survey results indicating a variety of workforce recruitment and retention collaborative relationships between schools and programs of public health and health departments, focus group participants only discussed examples regarding adjunct faculty. Focus groups did not identify any current formal partnerships to support other types of health department staffing efforts.

### FOCUS GROUP FINDINGS

**State Health Department** provided examples of health department staff serving as adjunct faculty members in the past, only one focus group participant was identified as a current adjunct faculty member.

**Mid-Sized Local Health Department** stated that many staff serve as adjunct professors at the academic institution with which they have an AHD partnership.

## STUDENT TRAINING

ASPPH-member schools and programs of public health partner, or rely, on governmental public health agencies to advance student training. According to 2023 survey results, student practicums/internships and guest lectures are the most common ways that ASPPH member institutions partner with state and local health departments. Of the 35 ASPPH member schools and programs of public health that participated in the 2023 survey, 100% noted at least one collaborative relationship with a state and/or local health department on student training efforts. Specifically, 100% indicated a practicum/internship partnership with a state and/or local health department, 89% collaborated on guest lectures with a health department(s), 46% on curriculum development, 34% on field trips, 29% on shadowing, 17% on apprenticeships, 17% on grand rounds, and only 14% on journal clubs.

## STUDENT TRAINING (CONTINUED)

### FOCUS GROUPS FINDINGS

All health departments interviewed in focus groups discussed practicums/internships and curriculum development partnerships, yet the specifics, breadth, and impact of these partnerships varied widely. Focus groups spent more time discussing practicum/internships and curriculum development partnerships than other types of student training partnerships, yet they did provide some insight into other collaborations.

### Practicums/Internships

**State Health Department** provides practicums/internships for students, yet it does not have a standard schedule or process for these partnerships, nor does it have a point person in charge of overseeing practicums/internships as a whole. In the past, a health department employee would contact a local academic institution and request students for specific projects, but this was an informal process that lacked continuity.

**Large Local Health Department** has a sophisticated internal structure for working with students and their academic institutions. The department currently has 41 formal affiliations with both in- and out-of-state academic institutions for internships and practicums, including opportunities for high school, undergraduate, graduate, and doctoral students. The local jurisdiction's contracts department and local-level officials take part in drafting and approving each affiliation agreement with academic institutions. The health department has a designated staff person in charge of practicums/internships who leads the application process, intern selection, onboarding, and communication with the academic institution throughout the practicum/internship period. The department provides monthly training sessions to all interns, and health department leaders meet with students to learn about their training experiences and identify challenges and solutions.

**Mid-Sized Local Health Department** expressed insufficient capacity to mentor/supervise students or to teach the necessary skills relevant to implementing health department projects.

## STUDENT TRAINING (CONTINUED)

### Curriculum Development

**State Health Department** is currently working with the state community college system to develop and launch an associate's degree in public health. The health department will help inform the degree curriculum to ensure it meets the training requirements codified in state statute for certain professions. The impetus behind this partnership is a desire to expand the pool of people interested in and prepared for governmental public health jobs with the hope that more people from underserved communities will enter the field. Regarding curriculum development with existing schools and programs of public health, the State Health Department is "finally" starting to have regular conversations about curriculum with these institutions as a result of CDC's Public Health Infrastructure Grant.

**Large Local Health Department** partners with select academic institutions to develop modules for students on ethics and legal ethics. This partnership grew from an identified need for students to be better prepared for professional workplace culture.

**Mid-Sized Local Health Department** employs "many" staff who sit on advisory committees and boards for the state medical school's public health program, including those focused on curriculum development.

### Other Student Training

**State Health Department** has formal agreements with two medical schools for fellows to visit the public health laboratory as part of their training, and local academic institutions bring students to the laboratory for tours on an ad hoc basis.

# BENEFITS OF BI-DIRECTIONAL RELATIONSHIPS

Successful bi-directional relationships emerge from mutual benefits, especially when it addresses problems that are challenging or impossible for a single organization to address (Kovach et al., 2019b). Relationships between governmental public health agencies and schools and programs of public health have the potential to improve public health practice, education, and research (Kovach et al., 2019a). Further, these relationships can also lead to improvements in health system innovations. Best practices in bi-directional relationships have demonstrated the significance of their far-reaching impact on students, communities, and health systems, and translation of evidence-based research to practice can transform health outcomes (Phillips et al., 2019).

A 2015 University of Illinois Chicago survey of 2,000 local health departments found that local health department officials and faculty from schools and programs of public health valued bi-directional relationships because they improve education and training, support public health accreditation, enhance local health department credibility, enhance local health department technological capabilities, and improve research and evidence-based practice. Local health department respondents reported a perceived increase in benefits as the degree of collaboration increased and found that local health department officials would like to collaborate more closely with schools and programs of public health (Kovach et al., 2019b). Some local health department officials have claimed that their health department's credibility was enhanced by collaborating with schools and programs of public health—demonstrating a relationship to funding organizations was important to highlight their access to "research and evaluation expertise" (Kovach et al., 2019b).

A 2017 study out of the University of Tennessee attempted to measure the extent to which U.S. local health departments are engaged in evidence-based public health and if this is influenced by the presence of an AHD partnership. Of the 376 valid responses received, there were 192 local health departments with formal AHD relationships, 80 with informal relationships, and 99 with no AHD relationships. Respondents with formal AHD relationships reported higher perceived organizational supports for evidence-based decision making and interventions compared with informal or no AHD relationships (Erwin et al., 2019).

On the other hand, schools and programs of public health find value in the ability of governmental public health agencies to provide students with opportunities to apply academic theory and classroom knowledge to real-world settings, to provide faculty and research staff with practice-based research opportunities to enhance skills and knowledge through workforce development, and to strengthen the evidence base of public health (Erwin et al., 2016).

> *"I'd say 15% to 20% of our partnerships in total are with academic institutions, because beyond what we do with the internship and fellowship programs, pretty much every division works with an academic institution in some capacity. Our STI programs will work with [a local college] and they'll host screening and testing events and education events. Or chronic disease has been working with a lot of academic institutions... We have academic institutions serving as evaluators on a lot of our grant-funded programs or just a lot of our grant projects."*
>
> *State Health Department*

BENEFITS OF BI-DIRECTIONAL RELATIONSHIPS

## TECHNICAL ASSISTANCE

Bi-directional relationships between governmental public health practice and academia improve collaborative research and evidence-based public health practice. While the evidence is thin for what works in public health practice, many of the limitations can be addressed through AHD-supported collaborative research that generates evidence for public health policy and practice (Erwin et al., 2016). A 2015 survey out of the University of Tennessee supports this claim with more than half of respondents stating that AHDs enhanced the successful completion of a CHA, a prerequisite for health department accreditation through PHAB (Erwin et al., 2016).

Evidence-based public health practice has become important as practitioners seek to enhance the efficient use of scarce resources (Erwin et al., 2016). Collaborative research occurring through AHDs provides opportunities to enhance the relevancy of research questions and study designs themselves, improves the potential for external validity because the real-world settings of public health practice are typically generalizable, and expands the possibilities for understanding barriers to implementation, including issues related to fidelity, when evidence-based activities or programs are scaled up (Erwin et al., 2016).

Successful bi-directional relationships in which both parties have a shared agenda and ongoing interaction can lead to increased probability that the research findings will be used in routine practice (Kilbourne & Atkins, 2014). As a result, AHDs can be an effective way to advance evidence-based decision-making and for implementing evidence-based programs and policies. This has both substantive and practical implications for public health practice and academia given the emphasis on evidence-based public health in the accreditation process. Health departments not yet participating in AHD partnerships need not ‘go it alone’; rather, they can draw from the extensive set of resources and tools from the AHD Learning Community supported by the Public Health Foundation (Erwin et al., 2019).

## TECHNICAL ASSISTANCE (CONTINUED)

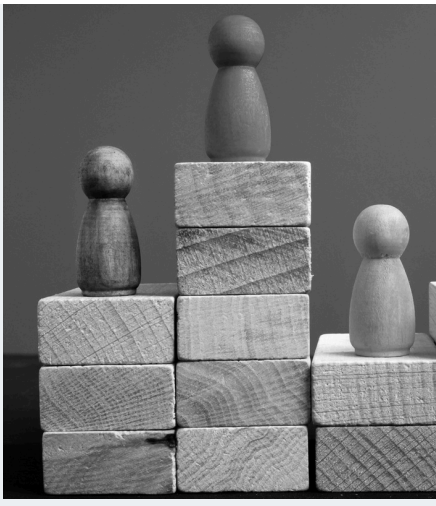
### FOCUS GROUP FINDINGS

**Large Local Health Department** benefits from academic institutions leading program evaluation because of their credibility as a neutral third party. The health department hopes to work with academic institutions to evaluate new initiatives from CDC's Public Health Infrastructure Grant regarding professional development and other recruitment and retention efforts.

**Mid-Sized Local Health Department** noted that working with its AHD partner on overall health department and division-specific strategy helps it stay accountable to its local board and general community.

*"I think what we've really found benefit with academic institutions is in the evaluations for some of our projects. Maybe because we are doing so many new initiatives under PHIG for recruitment and retention and professional development, having an outside party do the evaluation of those projects...would be helpful. Instead of us saying they're great, having a neutral third party."*

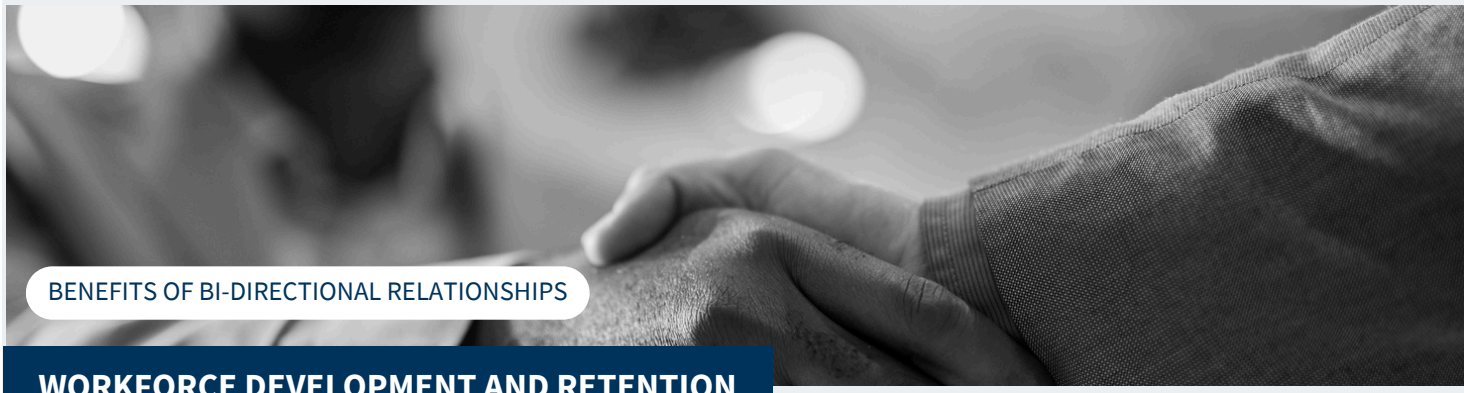
*State Health Department*



BENEFITS OF BI-DIRECTIONAL RELATIONSHIPS

## WORKFORCE DEVELOPMENT

A review of the literature did not produce findings related to the benefits of bi-directional relationships on workforce development efforts. While only one health department focus group briefly discussed a relationship in this area, no benefits were identified. This contrasts with the 89% of ASPPH member schools and programs of public health that noted a workforce development-related relationship in the 2023 ASPPH survey, highlighting a major gap in research in this area.



BENEFITS OF BI-DIRECTIONAL RELATIONSHIPS

## WORKFORCE DEVELOPMENT AND RETENTION

A review of the literature did not produce findings related to the benefits of bi-directional relationships on workforce recruitment and retention efforts. While health departments discussed their adjunct faculty relationships with schools and programs of public health, the conversations did not touch on the benefit of these relationships.

BENEFITS OF BI-DIRECTIONAL RELATIONSHIPS

## STUDENT TRAINING

Bi-directional relationships that foster student practicums/internships provide public health students with opportunities for practice-based learning. Governmental public health agencies, in turn, receive added capacity to complete projects outside of their employees' areas of expertise (Kovach et al., 2019b). A potential benefit of bi-directional relationships is students' preparation to enter the workforce, which may improve education and training opportunities for students and public health practitioners alike (Kovach et al., 2019b).

### FOCUS GROUP FINDINGS

**State Health Department** matches quality graduate-level students to identified department project needs making the practicum relationship beneficial for the health department. The health department has also had success in hiring former practicum students as permanent staff.

**Large Local Health Department,** with its 41 affiliation agreements and formal structure to onboard and manage practicum students, is able to match quality students to real department needs, thus increasing department capacity. The health department learns about new software and tools from these students and has hired approximately six former practicum students as permanent staff.

## STUDENT TRAINING (CONTINUED)

*"We've had a bunch of the interns coming through here, and we've given them real data sets, gotten some good visualizations from some of the data. It's an opportunity to have some hands-on work... For us, there are sometimes unique projects that you just temporarily need someone. They'll do that for us, and we also give them an opportunity to actually see the challenges. I think it's a great opportunity for them. It's just basically a practice partnership."*

*Large Local Health Department*

## *KNOWLEDGE GAP:*

*Neither the literature nor focus groups identified the benefits of bi-directional relationships on workforce development or workforce recruitment and retention.*

# CHALLENGES OF BI-DIRECTIONAL RELATIONSHIPS

While health departments are embracing bi-directional relationships with academia that are generally thought to be mutually beneficial, literature suggests that governmental public health agencies may have limited capacity to accomplish truly mutually beneficial relationships (Braund, 2019). By and large, the focus groups suggest that while health departments appreciate and want bi-directional relationships with academic institutions, barriers stemming from weaknesses in the bi-directional relationships or weaknesses at the health departments or academic institutions themselves often get in the way of truly symbiotic and impactful relationships.

CHALLENGES OF BI-DIRECTIONAL RELATIONSHIPS

## TECHNICAL ASSISTANCE

The University of Illinois Chicago survey of local health departments found several examples of local health departments and schools and programs of public health engaging in strategic initiatives; however, in many cases, the effectiveness of these initiatives remained unclear. For example, it was reported that schools and programs of public health frequently supported local health departments with CHA and CHIP activities. However, among local health departments that place importance on CHA and CHIP, 28% and 32% respectively did not think their collaboration with schools and programs of public health was effective. Reasons reported include lack of a long-term vision with initiatives that need to build both practitioners and faculty strategic and technical skills, such as informatics or systems thinking, and the need for practical ways to build evidence in practice or make evidence-based dissemination more relevant for practice (Kovach et al., 2019b).

**FOCUS GROUP FINDINGS:**

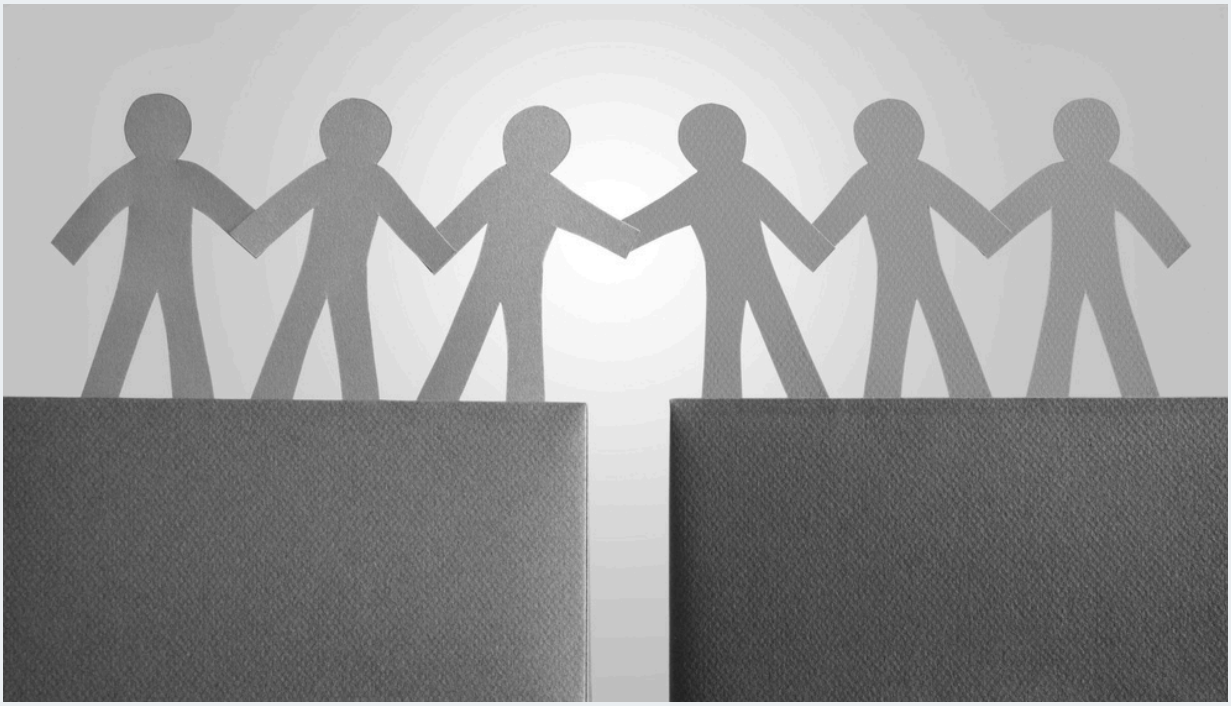
**Large Local Health Department** maintains a positive relationship with its AHD academic liaison; however, the liaison is stretched thin by competing priorities. As a result, the AHD is unable to fully deliver on its promised commitments related to technical assistance and other types of partnerships.

**Mid-Sized Local Health Department** finds value in having academic institutions serve as a neutral, third-party evaluator of health department programming. However, the health department must go through its state health department to contract with academic institutions in this way. As a result, the health department cannot work directly with its AHD academic partner. Challenges with this protocol include poor communication with the academic institution and inconsistent data. Further, staff voiced frustration with their lack of direct access to journal articles and library resources through their AHD partner institution.

CHALLENGES OF BI-DIRECTIONAL RELATIONSHIPS

## WORKFORCE DEVELOPMENT

A review of the literature did not produce findings related to the challenges of bi-directional relationships on workforce development efforts. While only one health department focus group briefly discussed a partnership in this area, no challenges were identified. This is in contrast to the 89% of ASPPH-member schools and programs of public health that noted a workforce development-related partnership in the 2023 ASPPH survey, highlighting a major gap in research in this area.



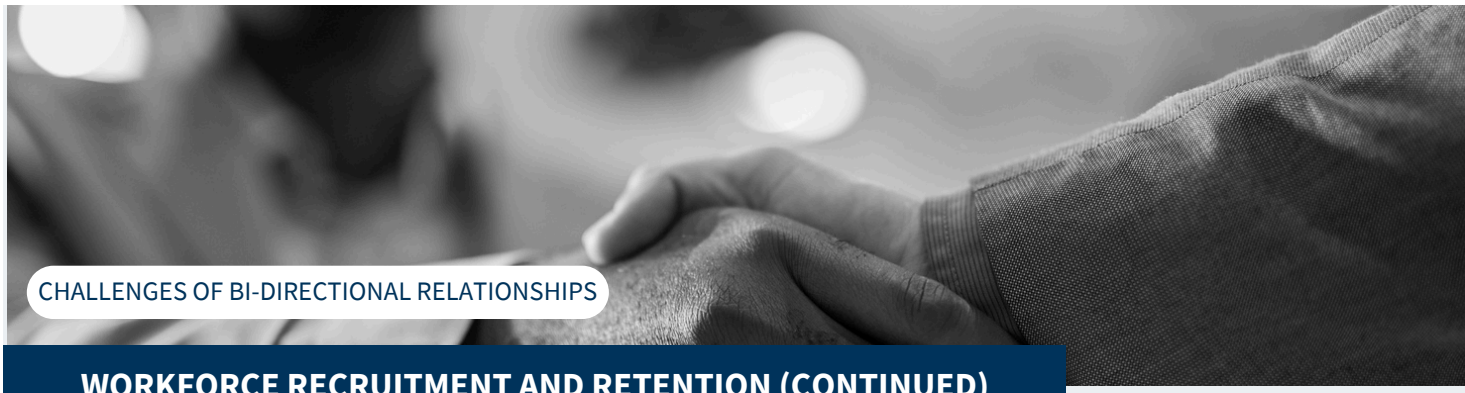
## WORKFORCE RECRUITMENT AND RETENTION

Leveraging bi-directional relationships for workforce recruitment and retention is a noted gap in the literature, which may be a result of some of the challenges addressed during the focus groups.

### FOCUS GROUP FINDINGS

**State Health Department** noted funding as a barrier to workforce recruitment and retention collaborations, including adjunct faculty appointments at academic institutions. This challenge was most evident with research institutions that often require faculty to bring in grant dollars. The sentiment among health department staff is that if health department staff are not an asset to the academic institution in this regard, they cannot teach. Similarly, if academic faculty do not have existing funding to work with the health department, it is very difficult for them to collaborate on practice projects.

**Mid-Sized Local Health Department** has confronted numerous challenges with its AHD partner institution regarding requests for staff time. Per the AHD agreement, no funding can be exchanged between the two partners. Even so, the AHD academic institution has previously requested that health department staff teach courses for free. Some of these requests came during periods in which health department staff were already serving on multiple advisory committees for the institution.





## WORKFORCE RECRUITMENT AND RETENTION (CONTINUED)

*"When we originally started talking about doing an academic health grant... the ask was that in order for there not to be money exchanged, they wanted me to teach a course as compensation for them giving us their partnership. I was the single epidemiologist at the time before the pandemic, and I was like, 'I don't have the capacity to teach. I don't have the capacity to do my job normally.' I think it is beneficial in a lot of ways, but there are other places where it's just like, why is it so hard?"*

*Mid-Sized Health Department*

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*"I have a very tangible issue with our academic health department partnership. I can't get journal articles effectively or efficiently... I should mention that myself and a number of my team members and people across the department are adjunct professors so that we can mentor their students and do all the things. For a while, we had gotten a special agreement. I could have access to the library to get articles... Now, if we wanted, we have to e-mail the person who's our key partner over there, to get her to get one of her staff to get us an article... It seems absolutely bonkers to me that as a partner, I cannot get journal articles... Just give me journal articles so I can actually do my job well."*

*Mid-Sized Health Department*

## STUDENT TRAINING

Local health department participants in the University of Illinois study spoke about challenges that arose from the physical distance between a local health department and schools and programs of public health, which was exacerbated when local health departments were located in small communities. Respondents spoke about long commutes and a lack of short-term housing as barriers to providing student learning activities. Proposed solutions for these barriers included providing students with stipends or paid internships, identifying free or affordable local housing, allowing students to telecommute, and seeking shared funding for collaborative projects (Kovach et al., 2019a).

### FOCUS GROUP FINDINGS

All health departments were understaffed at the time of the focus groups. Both the state health department and large local health department noted a significant number of job vacancies at entry-, mid-, and senior-level positions. The general sentiment was that academic institutions are an important part of the governmental public health workforce pipeline, ideally generating skilled, practice-ready graduates for jobs in health departments. While some health departments noted instances of hiring past practicum/internship students as permanent staff, there exist many weaknesses in the pipeline preventing graduates from entering the governmental public health workforce. These pipeline weaknesses are exacerbated by difficulty recruiting due to uncompetitive compensation packages.

## STUDENT TRAINING

### Internships and Practicum

**State Health Department** does not have a staff person managing the practicum/internship process. This makes it difficult to properly and easily identify potential projects, vet and match students to projects, and manage practicum/internship relationships with academic institutions. The health department also noted that incoming practicum students often lack experience in applied practice and lack structure and support from their respective academic institutions, putting the onus on health department staff to manage these students, familiarize them with practice, and teach them new skills.

### Curriculum Development

All health departments noted a disconnect between the knowledge and skills needed to work at a health department and the current curriculum of academic institutions. This disconnect is a major barrier to the health departments hiring graduates of these academic institutions. For example, two health departments noted that the public health curriculum of their local and/or state academic institutions is too general and does not prepare students for the specialized careers essential to fundamental health department functions such as epidemiology and informatics. One health department often hires graduates of out-of-state academic programs because the graduates from their state's school or program of public health do not have the requisite skills. This has proved to be a point of contention between the institutions. Health departments hypothesized reasons for this disconnect, including: a dearth of practice experience among academic faculty contributing to a lack of understanding regarding health department needs; a lack of health department capacity to educate academic institution faculty on health department needs and co-develop relevant curriculum; and academic institutions not placing a value or focus on training students for governmental public health, in part because of the need to attract incoming students.

## STUDENT TRAINING (CONTINUED)

### Other Student Training

**State Health Department** noted a major issue with hiring international graduates of schools and programs of public health. While a number of these international students have had the skills and experience the health department desires, visa requirements and the inability of a state agency to sponsor people through immigration processes have made it exceedingly difficult to hire these graduates as staff.

*"Having worked with our internship program for a few years, we get a student, but the student needs so much guidance and to be taught skills. I think that gap or the capacity issue comes in that we don't have time to teach them the skill set. We're here to offer the experience and share our knowledge."*

*Mid-Sized Health Department*

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*"We're trying through workforce development to make more standardized processes and more consistent processes [to identify and bring on interns] because these one-offs tend to benefit students who already have access. They know someone, their father's a doctor, they know the commissioner. They are tight with the professor, and so, the professor knows us and we get the call. We're in an effort to bring more equity into the availability of these processes. We're trying to make them more standardized rather than being one-offs."*

*State Health Department*

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*"At [academic institution] in particular, they're like, 'This is basically the only applied [practice] class that people would get.'...That was really the only opportunity that people have there. I don't know a lot about this, but what are they trying to do? I don't think that's necessarily a goal of that program. Their goal was not to create people that work in public health departments."*

*State Health Department*

## STUDENT TRAINING (CONTINUED)

*"I think sometimes academic organizations, they are so removed from the practice that you really need to lay it out for them. I think that's what we're going to be trying to do with this academic health department, is to really lay it out at what we need at the practitioner level and where the gaps are and how they can help fill them. I don't think they're going to know unless we tell them, because they can't guess what we do as practitioners... [But] I think it's time-consuming because everyone is so busy doing the job at hand and that's the lack of resource really because sometimes you've got to step away to get into that mindset of explaining what we need."*

*State Health Department*

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*"If I'm concerned about an epidemiologist, I would like the epidemiologist to have at least 12 credit hours in epidemiology and biostatistics instead of having a general epidemiologist who is just an MPH and doesn't have any quantitative or qualitative analysis skills... he workforce that we have is pretty robust, but we don't have the school of public health at this point which would offer courses in epidemiology, biostatistics, maternal child health, international health. Our school of public health is fairly new...The workforce that is coming from the universities is lacking at this point."*

*Large Local Health Department*

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*"One of the issues I perpetually see today from our small universities like [our local academic institution]... none of the professors have ever been in practice. It's a very academic, research-based lens. When the graduates come into our health department, they don't actually understand how to do the work on the ground. I think that's across the board."*

*Mid-Sized Health Department*

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*"In a perfect world, I would love to have our academic friends come in and really getting an idea of what is going on, on the day-to-day. I know that sounds silly and there's all sorts of logistic types of issues, but specifically walking in our shoes and understanding that we've got so many ropes and limitations around us that anytime we do something it's majorly affected."*

*Mid-Sized Health Department*

# OPPORTUNITIES TO STRENGTHEN BI-DIRECTIONAL RELATIONSHIPS

Through focus groups with health departments and a review of current literature, it is clear that public health practitioners and academicians alike recognize the benefit of bi-directional relationships between governmental public health agencies and schools and programs of public health. Bi-directional relationships have the potential to improve strategic planning, workforce development, workforce recruitment and retention, student training, and more, ultimately strengthening public health infrastructure and improving health outcomes across the country. Considering gaps in the literature and challenges described in focus groups, however, it is also evident that many unrealized opportunities exist for actors to support, build, and improve these relationships on a local, state, and national scale.

To achieve successful and sustained bi-directional relationships, governmental public health agencies and schools and programs of public health must develop relational foundations based in a shared vision, collaborative agenda, and integrated resources. This includes defining a formal or informal structure for partnership, including goals and objectives, roles and responsibilities, integration of resources, organizational capacity, and consistent communication. To realize the full benefit of these relationships, practice and academic partners must also consider the immediate needs of their institutions and individuals involved and long-term goals and desired outcomes. A number tools for developing and maintaining bi-directional relationships are available, including memoranda of understanding templates and AHD learning communities, made possible by the Public Health Foundation.

Bi-directional relationships between health departments and schools and programs of public health do not exist in a vacuum. While each relationship can be positively or negatively affected by policies or practices of the involved organizations, the larger ecosystem in which these institutions exist affects the partnership's success. As such, long-term investment in these partnerships will ensure that governmental public health agencies and schools and programs of public health have the necessary capacity and resources to successfully manage, operationalize, and maintain bi-directional relationships. Ongoing funding opportunities to support the development and sustainability of bi-directional relationships are needed to provide opportunities for a more long-term and strategic view for bi-directional relationships (Kovach et al., 2019b).

The review of the literature highlighted many gaps that, if filled, could significantly add to the evidence base for strengthening bi-directional relationships. For example, there is a dearth of literature regarding the existence of bi-directional relationships addressing workforce development and workforce recruitment and retention, let alone the benefits and challenges of these relationships. And while some literature identified examples of technical assistance and student training partnerships, the evidence of bi-directional relationship implementation, benefits, and challenges remains high-level and vague. To provide both governmental public health agencies and schools and programs of public health with the evidence base necessary to make informed decisions, confront challenges, and make the case for support, much more exploration into the nuances and operationalizing of bi-directional relationships is critical.

The lack of literature suggests that building successful bi-directional relationships has not been a priority of many local, state, and national public health actors. While national organizations that represent public health practice and academia have traditionally been aware of one another, they have yet to partner in significant ways to advance bi-directional relationships among their constituents. Yet bi-directional relationships between governmental public health agencies and academic institutions are essential for a resilient public health system.

These partnerships address immediate public health needs but also ensure mutual benefit, promote sustainability, and enhance infrastructure that is prepared to meet future public health challenges. We believe the Public Health Infrastructure Grant provides an exciting opportunity for national organizations to conduct joint information gathering, strategic planning, learning community facilitation, and tool and resource development to cultivate a truly supportive ecosystem for bi-directional relationships between governmental public health practice and academia.

# APPENDIX A

## BI-DIRECTIONAL RELATIONSHIPS FOCUS GROUP SCRIPT

ASPPH is the membership organization that represents accredited graduate schools and programs of public health. We are working with the National Network of Public Health Institutes to determine how academic public health can be a better partner to state, territorial, local, and tribal health departments and potentially provide proactive technical assistance in support of NNPHI and other national partners' PHIG funding.

As such, I must disclose that this work is supported by funds made available from the Centers for Disease Control and Prevention (CDC) of the U.S. Department of Health and Human Services (HHS), National Center for STLT Public Health Infrastructure and Workforce, through OE22-2203: Strengthening U.S. Public Health Infrastructure, Workforce, and Data Systems grant. The contents are those of the author(s) and do not necessarily represent the official views of, nor an endorsement, by CDC/HHS, or the U.S. Government.

Today I will be hosting a conversation with you all about your current, future, or desired formal and/or informal partnerships with academic institutions. I am happy to learn more about partnerships with other organizations as well if there are models that may be replicated with academia. I want to know what works and what doesn't work. While I will be recording today's session to transcribe and code for key themes, we will not publish any direct quotes without your consent. Do you all agree to be recorded today?

Thank you for your participation. I hope that this is a helpful step in sustaining or creating meaningful relationships with your colleagues in education for public health. We have reserved 90 minutes. Do you have any questions before we get started?

# APPENDIX A *(CONTINUED)*

## BI-DIRECTIONAL RELATIONSHIPS FOCUS GROUP SCRIPT

1. Can we please start with a round of introductions? I would like to know your name, title, and a little bit about what you do.
2. Can you tell me a bit about your health department? How are you structured? How many staff? What kind of work do you do?
3. Can you tell me about some of the things that your health department does really well?
4. Can you tell me about some of the challenges that you all are facing both as a health department and perhaps more individually in your role?
5. How does your health department define partnerships? If your health department doesn't define partnerships, how should you define partnerships?
6. How does or would this definition apply to formal vs. Informal partnerships?
7. Given these definitions, tell me about some of your partnerships? Which types of agencies or organizations are your partnering with?
8. What are the successes of those partnerships?
9. What are the challenges of those partnerships?

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# APPENDIX A *(CONTINUED)*

## BI-DIRECTIONAL RELATIONSHIPS FOCUS GROUP SCRIPT

10 Now that we have discussed partnerships a bit generally, let's drill into partnerships with academic institutions.

a. If academic partnerships have been discussed:

i. What do your partnerships with academic institutions look like?
ii. Are they formal or informal?
iii. What is the benefit of informal partnership?
iv. What is the benefit of formal partnership?
v. What is the process for formalizing partnerships?
vi. How are partnership managed/maintained on the health department end?
vii. What do you see as the benefits of partnering with academic institutions?
viii. What do you see as the challenges of partnering with academic institutions?
ix. How might academia address these challenges?
x. How does or how might your health department address these challenges?
xi. What do you wish academic institutions would do more of/less of?

Thank you for your candor as we discussed partnerships with academic institutions. Do you have any additional thoughts or questions for me? Any thoughts on what you would like to see come out of this work?

Again thank you, and I hope you don't mind if we need to reach out for additional clarity as we analyze the transcriptions and develop a report.

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# APPENDIX B

## *FOCUS GROUP ANALYSIS CODE BOOK*

Root code 1: Types of partnerships with academic institutions

- Child code A: Technical assistance
  - Facilitation, Strategic Planning, CHA/CHIP
  - Evaluation and assessment
  - Research and grants collaboration
  - Health department accreditation
  - Community outreach
- Child code B: Workforce development
  - Continuing education
  - Knowledge/skills training
  - Executive coaching
  - Leadership development
- Child code C: Workforce recruitment and retention
  - Shared staffing/Adjunct faculty
  - Surge capacity staffing
  - Joint search committee
  - Outsource of hiring
- Child code D: Student training
  - Practicum/internships/fellowships/ apprenticeships
  - Curriculum development/advising
  - Developing pipeline systems/structures
  - Guest lectures
  - Field trips
- Child code E: Other

Root code 2: Emerging partnerships

Root code 3: Current partnerships

- Child code A: Going well
- Child code B: Challenges

Root code 4: Not current partnerships

- Child code A: Needed/desired partnership
- Child code B: Past partnership
- Child code C: Challenges preventing partnerships

Root code 5: Partnership structure

Root code 6: Establishing partnerships

Root code 7: Maintaining partnerships

Root code 8: Academic Health Department

Root code 9: Mention of specific academic institutions

Root code 10: Successful relationships with non-academic partners

Root code 11: Great Quotes

Root code 12: Seems Important

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